

R.M.S. Titanic

Survival Analysis Report

A structured examination of passenger data to uncover patterns, correlations, and key influencing factors behind survival outcomes.

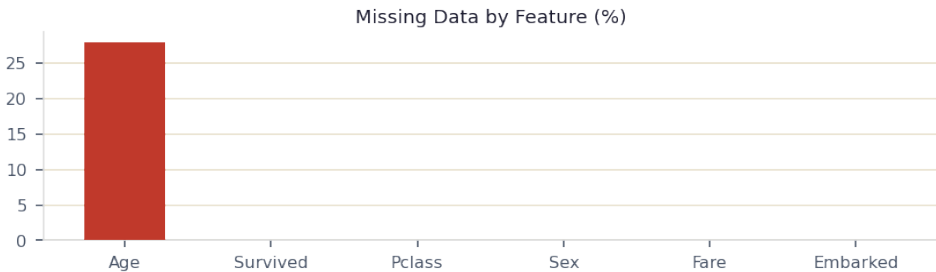
DATASET	VARIABLES	RECORDS	DATE
Titanic	10	50	June 2026

Dataset Overview

Summary of dataset dimensions, completeness, and headline survival statistics.

50 TOTAL PASSENGERS Sample analysed	22 SURVIVED 44.0% survival rate	28 PERISHED 56.0% of passengers
10 VARIABLES Mixed numeric/categorical	27.5 MEAN AGE Median: 27 yrs	27.90 MEAN FARE (GBP) Median: 15.75

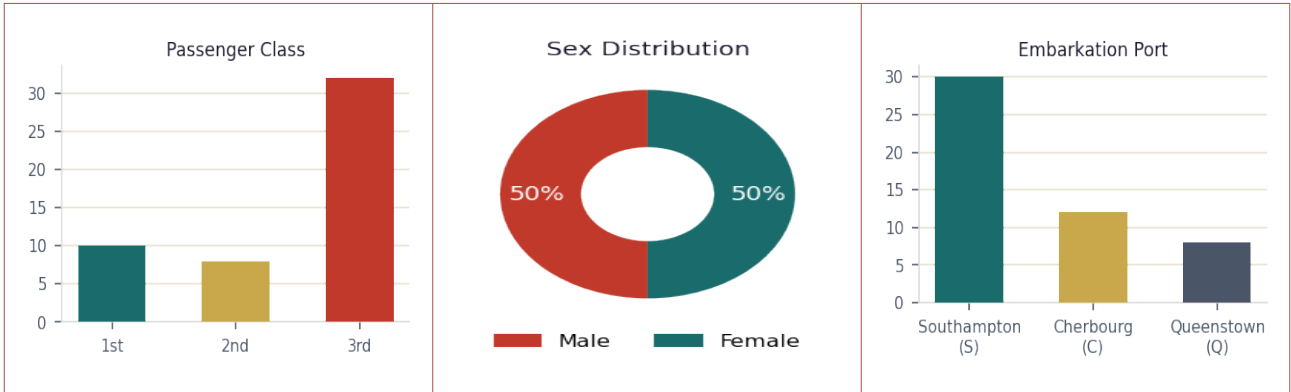
Missing Data Assessment

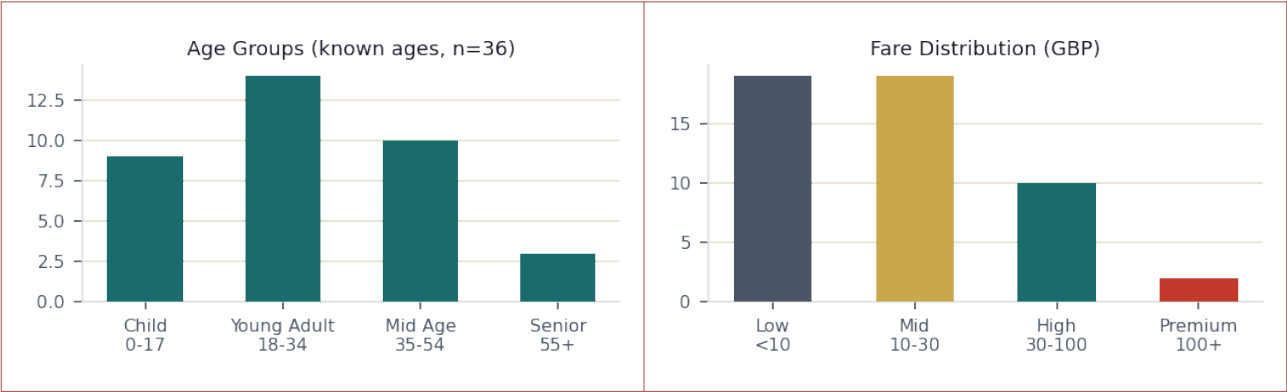


Age is the only field with missing values (14/50 records = 28%). Imputation — median fill or group-based — is recommended before predictive modelling.

Univariate Distributions

Individual feature distributions to understand shape, spread, and composition of each variable.



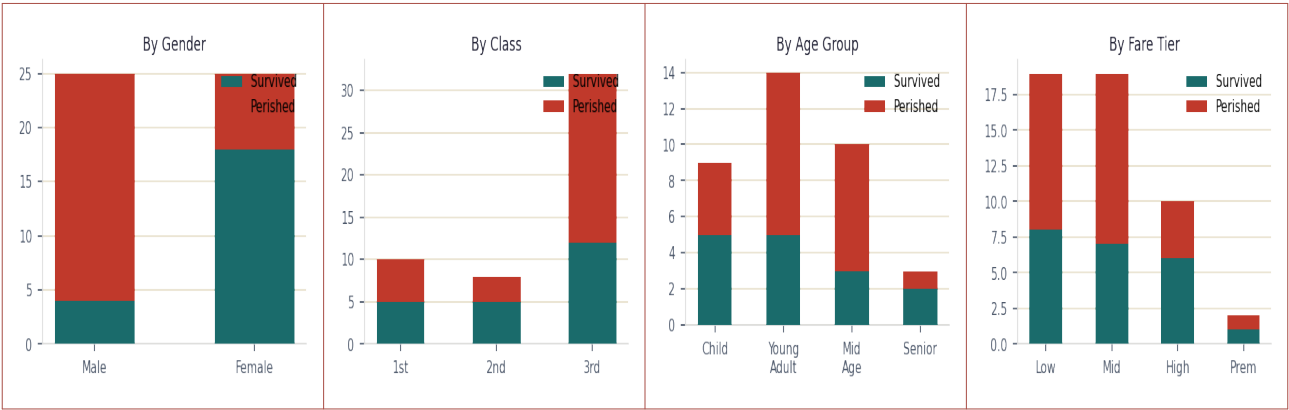


Survival Analysis

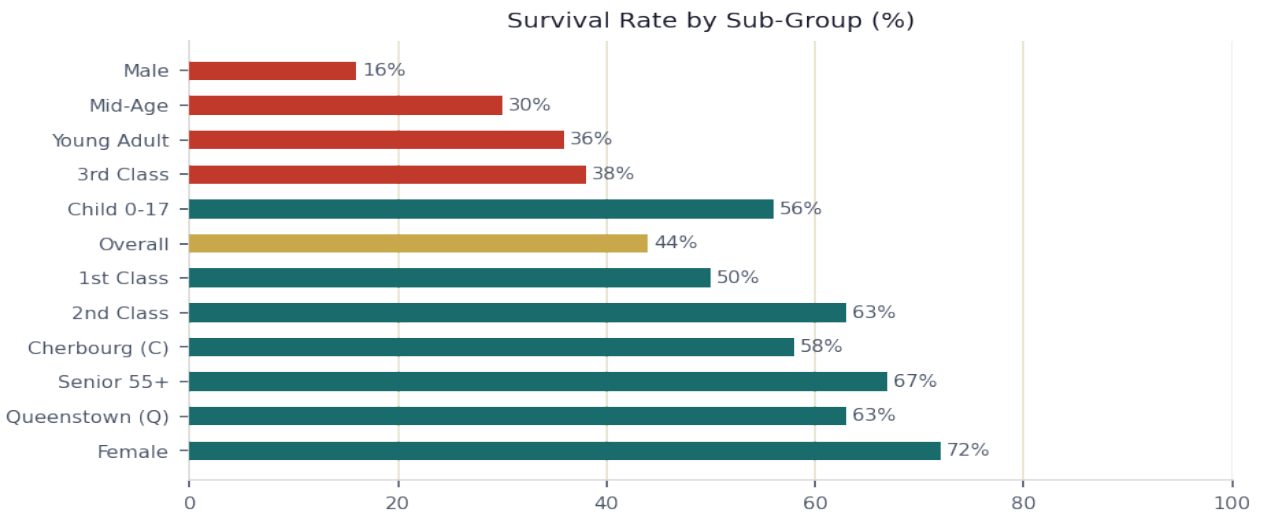
How survival rates differ across key attributes — identifying the most influential factors.

Survival Heatmap — Sex x Passenger Class

	1st Class	2nd Class	3rd Class
Female	4/5 80%	4/5 80%	10/15 67%
Male	1/5 20%	1/3 33%	2/17 12%

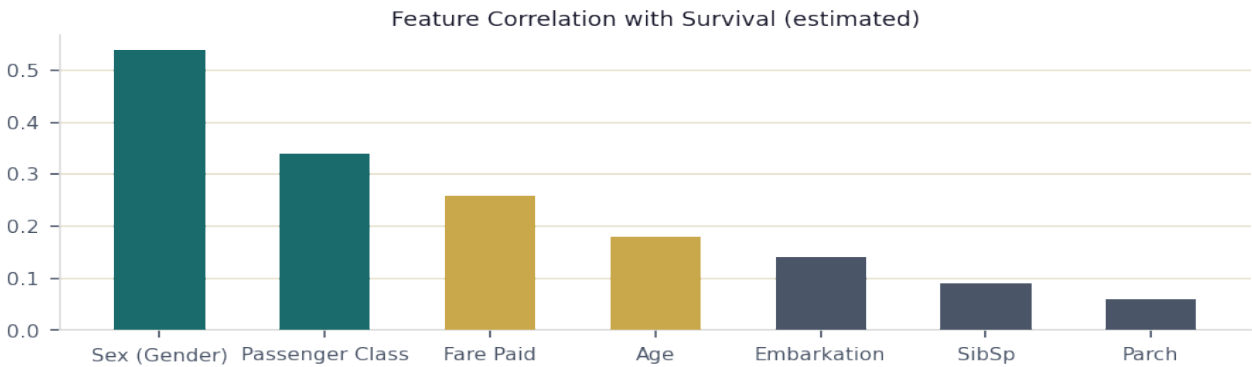


Survival Rate Comparison



Correlations & Key Influencing Factors

Estimated correlation strengths between passenger attributes and survival outcome.



Values estimated via point-biserial correlation (numeric) and Cramer's V (categorical). Sex has the strongest association followed by passenger class and fare paid.

FEATURE	TYPE	CORR.	STRENGTH	NOTES
Sex	Categorical	0.54	Strong	Female 72% vs male 16% — strongest predictor
Pclass	Ordinal	0.34	Moderate	Higher class correlates with better survival
Fare	Numeric	0.26	Moderate	Strong proxy for socioeconomic status
Age	Numeric	0.18	Weak	Children prioritised; 28% values missing
Embarked	Categorical	0.14	Weak	Cherbourg had more 1st-class passengers
SibSp	Numeric	0.09	Negligible	Siblings/spouses travelling
Parch	Numeric	0.06	Negligible	Parents or children aboard

Key Findings & Insights

- 1

Gender is the Strongest Survival Predictor

Female passengers survived at 72% vs only 16% for males — a 4.5x difference. The "women and children first" evacuation protocol is clearly reflected in the data, making sex the single most influential variable in this dataset.
- 2

Passenger Class Strongly Predicts Outcome

1st-class passengers survived at 50%, 2nd class at 63%, and 3rd class at 38%. Proximity to lifeboats and social privilege significantly determined who escaped. 3rd-class passengers formed 64% of this sample yet had the lowest survival rate.

3**Children Had Elevated Survival Chances**

Passengers aged under 18 survived at 56% — higher than young adults (36%) and middle-aged passengers (30%). This aligns with priority boarding policies. However the small sample (n=9) warrants caution in generalisation.

4**Higher Fares Correlate with Better Survival**

Passengers in the premium tier (100+ GBP) had 50% survival; the high tier (30-100 GBP) reached 60%. Fare is partly a proxy for class — wealthier passengers in 1st class were cabined closer to the upper decks and lifeboats.

5**28% of Age Records are Missing — Imputation Required**

Age is missing for 14 of 50 passengers. Since age correlates with survival, any predictive model must handle this carefully. Median imputation or group-based methods (by class and sex) are recommended before training machine learning models.

6**Cherbourg Passengers Showed Higher Survival**

Passengers embarking at Cherbourg survived at 58%, Southampton at 33%, and Queenstown at 63%. Cherbourg passengers were predominantly 1st class, suggesting embarkation port is a proxy for wealth rather than an independent survival factor.

Sample Data Preview

First 20 rows of the dataset for reference and validation.

#	Name	Class	Sex	Age	Fare	Embarked	Outcome
1	Braund, Mr. Owen Harris	3	Male	22	7.25	S	Perished
2	Cumings, Mrs. John Bradley	1	Female	38	71.28	C	Survived
3	Heikkinen, Miss. Laina	3	Female	26	7.92	S	Survived
4	Futrelle, Mrs. Jacques Heath	1	Female	35	53.10	S	Survived
5	Allen, Mr. William Henry	3	Male	35	8.05	S	Perished
6	Moran, Mr. James	3	Male	N/A	8.46	Q	Perished
7	McCarthy, Mr. Timothy J	1	Male	54	51.86	S	Perished
8	Palsson, Master. Gosta Leonard	3	Male	2	21.08	S	Perished
9	Johnson, Mrs. Oscar W	3	Female	27	11.13	S	Survived
10	Nasser, Mrs. Nicholas	2	Female	14	30.07	C	Survived
11	Sandstrom, Miss. Marguerite Rut	3	Female	4	16.70	S	Survived
12	Bonnell, Miss. Elizabeth	1	Female	58	26.55	S	Survived
13	Saunderscock, Mr. William Henry	3	Male	20	8.05	S	Perished
14	Andersson, Mr. Anders Johan	3	Male	39	31.28	S	Perished
15	Vestrom, Miss. Hulda Amanda	3	Female	14	7.85	S	Perished
16	Hewlett, Mrs. (Mary D Kingcome)	2	Female	55	16.00	S	Survived
17	Rice, Master. Eugene	3	Male	2	29.13	Q	Perished

18	Williams, Mr. Charles Eugene	2	Male	N/A	13.00	S	Survived
19	Vander Planke, Mrs. Julius	3	Female	31	18.00	S	Perished
20	Masselmani, Mrs. Fatima	3	Female	N/A	7.23	C	Survived